Multimodal Embedding Disentanglement

Laura Montesdeoca Fenoy
mont@itu.dk

Abstract

We present a systematic study of disentangled multimodal representation learning using the FEIDEGGER dataset, which contains multiple textual annotations per image. Our approach builds on the CLIP-ViT-B-32-multilingual-v1 architecture, finetuned with a loss function which encourages the separation of content and subjective information in the shared embedding space. Our results show a modest improvement in text-to-image retrieval (R@1 0.112 vs. 0.106 for standard finetuning). Critically, we demonstrate that disentanglement dramatically improves embedding consistency for semantically equivalent descriptions, reducing intra-sample distances by 48% compared to baseline CLIP and 84% compared to standard finetuning. The code for this project is available on GitHub at github.com/laumonfe/embeddingDisentanglement.

Figure 1: Example of embedding clustering for five different annotations of the same dress. The original sentences "Ein locker geschnittenes Mini-Wickelkleid in schwarz mit einem Blumenmuster in weiß, grau und pink. Das Kleid hat kurze Ärmel und einen tiefen V-Ausschnitt." corresponds to point number 5, "schwarzes kurzes Klied mit weißen blumen als Aufdruck es hat kurze Ärmel uund eine n V ausschnizt" is point number 3. They have a cosine similarity of 0.6097

1 Introduction

Real-world information is inherently multimodal, encompassing diverse forms such as text, images, and audio. However, obtaining meaningful representations which capture the richness of the data and enable a seamless search experience is a non-trivial task. While recent advances in multimodal learning have focused on unified embedding spaces (Chen et al., 2025; He et al., 2025),the notion of disentangled embeddings in the context of cross-modal retrieval remains an underexplored yet promising research direction. In particular, when multiple textual descriptions exist for the same image, standard embedding methods produce representations that vary not only due to genuine semantic differences but also due to stylistic choices, subjective interpretations, and linguistic variations.

Consider a fashion image that may be described as "A loosely cut mini wrap dress in black with a floral pattern in white, gray, and pink. The dress has short sleeves and a deep V-neck." by one annotator and "Black short dress with a white floral print. It has short sleeves and a V-neck" by another. While both descriptions refer to the same visual content, the dress in Figure 1, their embeddings in standard multimodal spaces (in this case using clip-ViT-B-32-multilingual-v1) can differ substantially due to linguistic style rather than semantic content. This entanglement of content and style poses challenges for retrieval systems, where we ideally want descriptions of the same image to cluster tightly together regardless of how they are expressed.

By separating visual content from linguistic style in the shared embedding space, we hypothesize that we can achieve more robust retrieval.

Our contributions are twofold:

1. We systematically evaluate the effect of disentanglement by explicitly separating content from style in the embedding space.

2. We provide empirical evidence that disentanglement significantly improves representation quality, achieving 48% lower intra-sample variance compared to baseline CLIP

while maintaining competitive retrieval performance.

1.1 Research Question

Does disentangling visual content from linguistic style lead to more robust and interpretable image-text retrieval compared to standard unified embeddings?

2 Related Work

Recent advances in vision-language models have demonstrated the effectiveness of learning unified embedding spaces that align visual and textual modalities. CLIP (Radford et al., 2021) pioneered contrastive pretraining on large-scale image-text pairs, enabling zero-shot transfer across diverse tasks. Subsequent work has extended this paradigm to multilingual settings (Chen et al., 2023) and explored architectural improvements. In the fashion domain specifically, FashionBERT (Gao et al., 2020) achieves strong performance on fashion-specific cross-modal retrieval. However, these models typically learn entangled representations where semantic content and stylistic attributes are intermixed, limiting their interpretability and fine-grained controllability.

Disentanglement has been extensively studied as a representation learning principle in generative models in the vision domain, (Chen et al., 2016; Higgins et al., 2017; Karras et al., 2019; Yin et al., 2019; Nie et al., 2020; Cai et al., 2023), where the goal is to separate independent factors of variation in the latent space, which allows for more control.

In the context of natural language processing, disentanglement has been explored primarily for text style transfer and controllable generation. Works like (Vishnubhotla et al., 2021) conduct a systematic evaluation of disentangled representation learning for separating content (semantics) from form (style) in text, demonstrating that current VAE-based approaches require substantial supervision to achieve meaningful separation. Their findings highlight the challenges of applying disentanglement to discrete, high-dimensional text data. Similarly, (Liao et al., 2020) explore disentangled word embeddings to improve interpretability. These works underscore the difficulty of learning fine-grained disentanglement without strong supervisory signals, a challenge that extends to the multimodal setting.

3 Data

For this task, we employ the FEIDEGGER dataset from Zalando (Lefakis et al., 2018). The dataset consists of 8,732 dress images, each paired with approximately five captions in German written by different annotators. This diversity introduces significant intra-class variation across multiple dimensions: some annotators favor objective, catalogue-style descriptions focusing on concrete visual attributes (e.g., "ein schwarzes kleid ohne ärmel und was bis fast zu den knien reicht." - a black dress without sleeves that reaches almost to the knees), while others adopt a more subjective perspective emphasizing use cases and aesthetic qualities (e.g., "sommer, kurze, sexy, elegantes, party, schwarze kleid" - summer, short, sexy, elegant, party, black dress).

Beyond content choice, annotators differ systematically in their linguistic patterns. Some use concise descriptions, while others prefer elaborate and rich sentences. Certain annotators employ fashion-specific terminology, while others use everyday language. This stylistic heterogeneity is not random noise but rather reflects genuine individual differences in how people perceive and describe fashion items and, ideally, should not affect retrieval of the same underlying garment.

This rich annotator diversity makes FEIDEGGER an ideal benchmark for studying disentangled multimodal representations. Unlike datasets with single canonical captions or synthetic paraphrases, FEIDEGGER captures authentic human variation in how the same visual content can be described. By learning to separate content-invariant features from annotator-specific stylistic choices, our model can achieve more consistent embeddings that reflect what is shown rather than how it is described.

4 Methodology

This section describes our approach, including how we generated and prepared the data, as well as the design of our model.

4.1 Data Generation

The dataset is partitioned into training (80%), validation (10%), and test (10%) sets, with splits performed at the item level to ensure all descriptions of the same image remain in the same partition. Training and validation are used during the finetunning of the model. And test is used during the performance evaluation of the different models. During the training phase, we utilize two distinct dataset formats:

- **CLIPDataset** treats each image-caption pair independently, constructing batches where every sample consists of a single image and its corresponding textual description. This format is considered standard in the original CLIP implementation (Radford et al., 2021), where the objective is to align individual image and text embeddings.
- **GroupedCLIPDataset** organizes data such that all captions associated with the same image are grouped together within a batch. This enables multi-label supervision, allowing the model to consider multiple positive text matches for each image.

4.2 Data Preprocessing

We follow the standard data preprocessing pipeline used in CLIP. For the text encoder, we tokenize all text using the CLIP tokenizer, which lowercases, strips accents, and truncates or pads sequences to a fixed length. For the vision encoder, all images are resized and center-cropped to 224×224 pixels, converted to RGB, and normalized using the mean and standard deviation values from the original CLIP training[1]. This ensures compatibility with the pretrained models and consistent input distributions for all models.

4.3 Model Architecture

The core architecture in this work is based on the **clip-ViT-B-32-multilingual-v1** model from the HuggingFace Sentence Transformers library[2]. This architecture follows the dual-encoder paradigm of CLIP, consisting of two main components:

- **Image Encoder:** The image encoder is implemented using the **CLIPVisionModel** from the HuggingFace Transformers library, which is a Vision Transformer (ViT-B/32) backbone. This encoder processes input images and produces fixed-length visual embeddings.
- **Text Encoder:** The text encoder is based on **DistilBertModel**, also from the Transformers library. Text inputs are tokenized and passed through DistilBERT, to obtain text embeddings in the same shared space as the image embeddings.

Both encoders are initialized from pretrained weights, and their architectures remain unchanged throughout our experiments. Instead, our work focuses on exploring disentanglement in the shared embedding space by modifying the loss function during training.

4.4 Model Kinds

To provide a comprehensive evaluation of our approach, we include three types of models in our experiments: baseline, finetuned, and disentangled.

- **Baseline:** The baseline model serves as a reference point, using pretrained weights without any additional finetuning or disentanglement objectives. This allows us to measure the performance of standard, off-the-shelf multimodal representations on our retrieval task.
- **Finetuned:** The finetuned model is trained on our dataset using a symmetric cross entropy loss, which the standard loss function in CLIP (Radford et al., 2021) . This model serves as a strong baseline, allowing us to fairly assess whether disentanglement provides additional benefits beyond those gained by simply finetuning the pretrained representations on our data.
- **Disentangled:** The disentangled model is also finetuned on the FEIDEGGER Dataset, but with a loss function that explicitly encourages the separation of content and subjective information in the embedding space, further details in Section 4.5 .

During training, we evaluate each model variant (finetuned and disentangled) using both data generation methods described in Section 4.1: the *CLIPDataset* format (single caption per image, which we will refer to as *Default* from now on) and the *GroupedCLIPDataset* format (multiple captions per image, which will be refered to as *Grouped*). This results in five distinct models: **Baseline, Finetuned Default, Finetuned Grouped, Disentangled Default, and Disentangled Grouped**.

By systematically evaluating each model under both data generation strategies, we are able to isolate the effects of finetuning, disentanglement, and data organization.

[1] https://huggingface.co/docs/transformers/en/model_doc/clip

[2] https://huggingface.co/sentence-transformers/clip-ViT-B-32-multilingual-v1

4.5 Loss Functions

We explore several loss functions to train and analyze the behavior of our multimodal embedding models. Each loss is designed to leverage different aspects of the data and to encourage specific properties in the learned representations.

Contrastive Loss. The standard contrastive loss aligns image and text embeddings in a shared space. For a batch of N image-text pairs, we compute the cosine similarity between all image and text embeddings, normalized by a temperature parameter τ. The loss encourages matching pairs to have high similarity and non-matching pairs to have low similarity:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{2N}\sum_{i=1}^{N}\left(\log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}} + \log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ji}/\tau}}\right) \tag{1}$$

where $s_{ij} = \mathbf{i}_i^\top \mathbf{t}_j$ is the cosine similarity between normalized image embedding $\mathbf{i}_i$ and text embedding $\mathbf{t}_j$.

Grouped Contrastive Loss. To leverage multiple captions per image, we use a multi-positive contrastive loss. For each image, all its associated captions are treated as positives, and all other captions as negatives. The loss for each image is:

$$\mathcal{L}_{\text{contrastive-grouped}} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{\sum_{j\in\mathcal{P}_i} e^{s_{ij}/\tau}}{\sum_{k=1}^{M} e^{s_{ik}/\tau}}\right) \tag{2}$$

where N is the number of images, M is the total number of captions, $\mathcal{P}_i$ is the set of positive caption indices for image i, s_{ij} is the cosine similarity between image i and caption j, and τ is the temperature parameter.

Disentangled Loss. To encourage separation of content and subjective information, we split each embedding into two halves along the feature dimension: a content subspace and a subjective subspace. Formally, given an embedding $\mathbf{z} \in \mathbb{R}^d$, we partition it as $\mathbf{z} = [\mathbf{z}_c; \mathbf{z}_s]$, where $\mathbf{z}_c \in \mathbb{R}^{d/2}$ represents content-related features and $\mathbf{z}_s \in \mathbb{R}^{d/2}$ captures subjective or stylistic attributes. The disentanglement objective consists of three complementary terms:

- **Content alignment:** A contrastive loss applied to the content subspace, aligning images and texts based on objective visual content. This uses the same formulation as Equation 1, but applied only to the content embeddings $\mathbf{i}_c$ and $\mathbf{t}_c$.
- **Cross-modal subjective independence:** $\mathcal{L}_{\text{orth-cross}} = \mathcal{L}_{\text{orth}}(\mathbf{i}_s, \mathbf{t}_s)$ penalizes correlation between the subjective parts of image and text embeddings, encouraging subjective information to remain modality-specific.

  To enforce independence between different subspaces, we use an orthogonality penalty defined as:

  $$\mathcal{L}_{\text{orth}}(\mathbf{a}, \mathbf{b}) = \frac{1}{N}\sum_{i=1}^{N}\left|\mathbf{a}_i^\top \mathbf{b}_i\right| \tag{3}$$

  where $|\cdot|$ denotes the absolute value of the dot product between normalized embeddings. This is minimized when the two subspaces are orthogonal.
- **Intra-modal disentanglement:** $\mathcal{L}_{\text{orth-intra}} = \mathcal{L}_{\text{orth}}(\mathbf{t}_c, \mathbf{t}_s)$ penalizes correlation between content and subjective parts within text embeddings, ensuring that stylistic choices do not leak into content representations. It is also enforced though the orthogonalitry penalty defined in Equation 3.

The total disentangled loss is a weighted combination of these three terms:

$$\mathcal{L}_{\text{disentangled}} = \alpha\,\mathcal{L}_{\text{content}} + \beta\,\mathcal{L}_{\text{orth-cross}} + \gamma\,\mathcal{L}_{\text{orth-intra}} \tag{4}$$

where α, β, and γ are hyperparameters that control the relative importance of each objective. In our experiments, we set $\alpha = 1.0$, $\beta = 1.0$, and $\gamma = 0.1$ to prioritize content alignment while regularizing for disentanglement.

Grouped Disentangled Loss. This loss extends the disentangled loss to the grouped setting, using the multi-positive contrastive loss for the content part and expanding the subjective penalty to all caption-image pairs within a group. The subjective part of each image embedding is duplicated to match the number of associated captions, and the orthogonality penalties are computed accordingly.

5 Results

In the following section we present both quantitative and qualitative results to comprehensively evaluate the performance and characteristics of our models.

5.1 Quantitative Results

To assess the quality of the learned multimodal embeddings, we evaluate all models on a retrieval task using the FEIDEGGER dataset. Specifically, we measure how well the model can retrieve the correct image given a text query (text-to-image retrieval). For each model variant, we compute the standard retrieval metrics on the test split.

5.1.1 Retrieval Metrics

We report $\mathrm{Recall@}K$ for $K \in \{1, 5, 10, 50\}$, as well as the mean rank of the ground truth image for each text query. $\mathrm{Recall@}K$ measures the proportion of queries for which at least one correct image appears in the top-K retrieved results. The mean ground truth rank provides a complementary view of retrieval quality.

Model	Type	R@1	R@5	R@10	R@50
Baseline		0.021	0.025	0.039	0.136
Finetuned	Default	0.106	0.118	0.202	0.495
Finetuned	Grouped	0.051	0.060	0.103	0.325
Disentangled	Default	**0.112**	**0.124**	**0.206**	**0.507**
Disentangled	Grouped	0.063	0.075	0.131	0.384

Table 1: Recall@K for text-to-image retrieval. Higher values indicate better performance, with Recall@K measuring the proportion of queries where the correct image appears in the top K results.

As shown in Table 1, both the disentangled and finetuned models outperform the baseline across all retrieval metrics, regardless of the data generation type. These results underscore the critical importance of domain-specific finetuning for improving retrieval performance on specialized datasets.

For both the disentangled and finetuned models, the default (single-caption) setting consistently outperforms the grouped (multi-caption) setting. For instance, Recall@1 drops from 0.112 (default) to 0.063 (grouped) for the disentangled model, and from 0.106 to 0.051 for the finetuned model. Similarly, mean ground truth rank increases in the grouped setting, as shown in Table 2. This suggests that, in this setup, grouping multiple captions per image as positives may introduce ambiguity or conflicting supervision, making the retrieval task harder for the model.

The disentangled and finetuned models perform similarly in the default setting, with the disentangled model slightly ahead in Recall@1 and mean rank. This indicates that the disentanglement loss does not harm retrieval performance and may even provide a slight benefit, possibly by encouraging more structured representations.

Model	Type	Mean GT Rank
Baseline		675.6
Finetuned	Default	155.1
Finetuned	Grouped	300.5
Disentangled	Default	**148.0**
Disentangled	Grouped	244.2

Table 2: Mean ground truth rank for each model and dataset type. Lower ranks indicate better retrieval performance, with values representing the average position of the correct image among 880 candidates in text-to-image retrieval.

5.1.2 Intra-Sample Distance Analysis

To quantify the effectiveness of our disentanglement approach, we measured the consistency of text embeddings for different descriptions of the same image using three complementary metrics: average pairwise distance, maximum pairwise distance, and mean distance to centroid. Lower values indicate that embeddings for the same item are more tightly clustered, suggesting successful disentanglement of style from content.

Table 3 presents the intra-sample distance metrics across all model variants. The disentangled model with default data achieves substantially lower distances across all metrics compared to baseline and finetuned alternatives. Specifically, the average pairwise distance is 0.0497, representing a 48% improvement over the baseline (0.0965) and an 84% improvement over the finetuned model with default data (0.3094). These results demonstrate that our disentanglement objective successfully encourages the model to generate similar embeddings for semantically equivalent descriptions, regardless of stylistic variations.

Figure 2 corroborates this finding by visualizing the embedding space for the same example item from Figure 1, now using the Disentangled Default model. The embeddings are substantially more compact than those produced by the Baseline model.

5.2 Qualitative Results

To complement the quantitative analysis, we present qualitative visualizations of the embedding spaces and retrieval results in Appendices B and C. These examples illustrate how different model variants handle semantic similarity and embedding consistency across various fashion items.

Model	Data Type	Avg. Pairwise Distance	Max. Pairwise Distance	Mean Centroid Distance
Baseline		0.0965 ± 0.0361	0.1547 ± 0.0610	0.0384 ± 0.0145
Finetuned	Default	0.3094 ± 0.1551	0.5480 ± 0.2548	0.1306 ± 0.0693
Finetuned	Grouped	0.3525 ± 0.2078	0.6555 ± 0.3402	0.1529 ± 0.0967
Disentangled	Default	**0.0497 ± 0.0332**	**0.0935 ± 0.0633**	**0.0197 ± 0.0131**
Disentangled	Grouped	0.2638 ± 0.1406	0.4841 ± 0.2336	0.1104 ± 0.0612

Table 3: Intra-sample distance metrics for text embeddings across different models. Lower values indicate more compact and consistent embeddings for different descriptions of the same image.

Figure 2: Example of embedding clustering for five different annotations of the same dress. The embeddings have been calculated with the Disentangled Default model. The cosine similarity between point number 5 and point number 3 is 0.9229

Retrieval Results (Appendix B) The retrieval visualizations demonstrate how each model ranks images in response to text queries.

Embedding Space Visualizations (Appendix C) We visualize the embedding spaces for five representative items using dimensionality reduction. Each example shows how multiple text descriptions of the same garment are positioned in the learned embedding space across different model variants.

Overall, the qualitative analysis confirms that the disentangled default model achieves the best balance between embedding consistency and semantic fidelity, supporting our quantitative findings in Section 5.1.

6 Conclusion

In this work, we systematically explored disentangled multimodal representation learning using the FEIDEGGER dataset, leveraging a CLIP-based dual-encoder architecture with both vision and text backbones. We introduced and compared several loss functions, including standard and grouped contrastive losses, as well as novel disentanglement objectives designed to separate content and subjective information within the shared embedding space.

Our experimental results demonstrate that both finetuning and disentanglement strategies substantially improve retrieval performance over the baseline. Notably, the disentangled model achieves the best overall results in the default (single-caption) setting, indicating that encouraging structured representations does not compromise, and may even enhance, retrieval quality. However, we observe that grouped (multi-caption) supervision, as implemented here, does not yield further improvements and may introduce ambiguity that hinders performance.

These findings suggest that while disentanglement is a promising direction for interpretable and robust multimodal learning, the design of supervision and loss functions remains critical. Future work may explore more sophisticated grouping strategies, alternative disentanglement objectives, and applications to other multimodal tasks and datasets.

References

Yufei Cai, Yuxiang Wei, Zhilong Ji, Jinfeng Bai, Hu Han, and Wangmeng Zuo. 2023. Decoupled textual embeddings for customized image generation. *Preprint*, arXiv:2312.11826.

Guanhua Chen, Lu Hou, Yun Chen, Wenliang Dai, Lifeng Shang, Xin Jiang, Qun Liu, Jia Pan, and Wenping Wang. 2023. mCLIP: Multilingual CLIP via cross-lingual transfer. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13028–13043, Toronto, Canada. Association for Computational Linguistics.

Jiuhai Chen, Zhiyang Xu, Xichen Pan, Yushi Hu, Can Qin, Tom Goldstein, Lifu Huang, Tianyi Zhou, Saining Xie, Silvio Savarese, Le Xue, Caiming Xiong, and Ran Xu. 2025. Blip3-o: A family of fully open unified multimodal models-architecture, training and dataset. *Preprint*, arXiv:2505.09568.

Xi Chen, Yan Duan, Rein Houthooft, John Schulman, Ilya Sutskever, and Pieter Abbeel. 2016. Infogan: Interpretable representation learning by information maximizing generative adversarial nets. *Preprint*, arXiv:1606.03657.

Dehong Gao, Linbo Jin, Ben Chen, Minghui Qiu, Peng Li, Yi Wei, Yi Hu, and Hao Wang. 2020. Fashionbert: Text and image matching with adaptive loss for cross-modal retrieval. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '20, page 2251–2260, New York, NY, USA. Association for Computing Machinery.

Yufei He, Yuan Sui, Xiaoxin He, Yue Liu, Yifei Sun, and Bryan Hooi. 2025. Unigraph2: Learning a unified embedding space to bind multimodal graphs. *Preprint*, arXiv:2502.00806.

Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, and Alexander Lerchner. 2017. beta-VAE: Learning basic visual concepts with a constrained variational framework. In *International Conference on Learning Representations*.

Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. *Preprint*, arXiv:1812.04948.

Leonidas Lefakis, Alan Akbik, and Roland Vollgraf. 2018. Feidegger: A multi-modal corpus of fashion images and descriptions in german. In *LREC 2018, 11th Language Resources and Evaluation Conference*.

Keng-Te Liao, Cheng-Syuan Lee, Zhong-Yu Huang, and Shou-de Lin. 2020. Explaining word embeddings via disentangled representation. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, pages 720–725, Suzhou, China. Association for Computational Linguistics.

Weili Nie, Tero Karras, Animesh Garg, Shoubhik Debnath, Anjul Patney, Ankit B. Patel, and Anima Anandkumar. 2020. Semi-supervised stylegan for disentanglement learning. *Preprint*, arXiv:2003.03461.

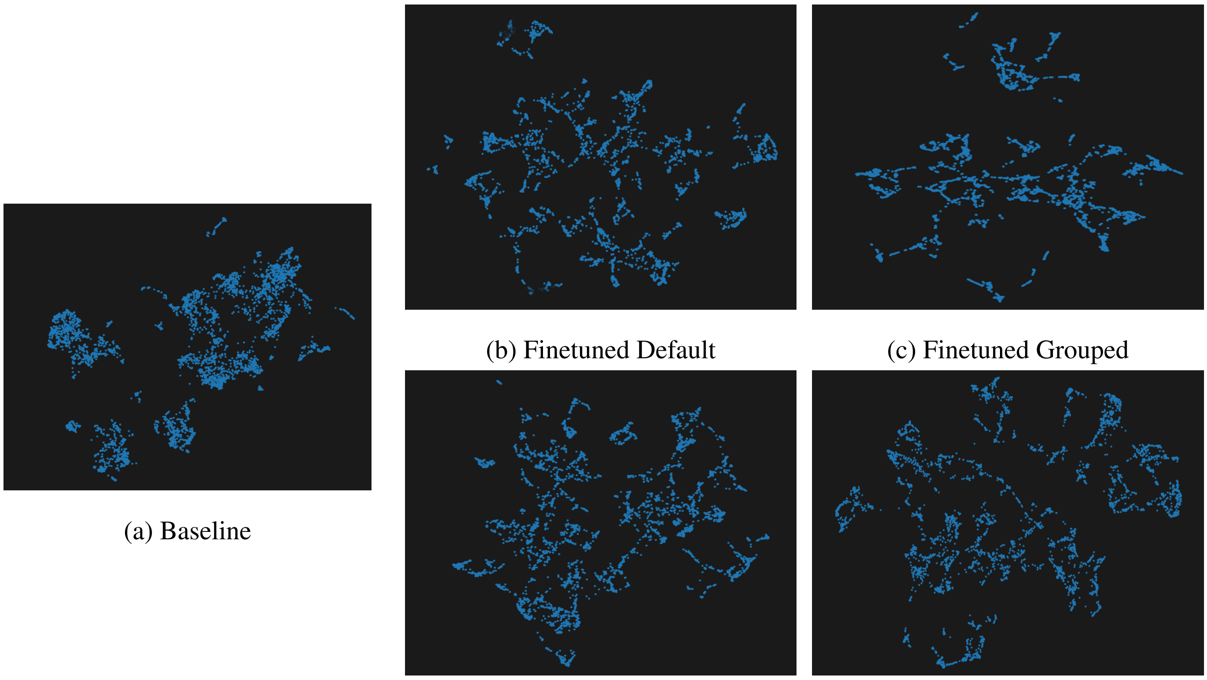
Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. *Preprint*, arXiv:2103.00020.

Krishnapriya Vishnubhotla, Graeme Hirst, and Frank Rudzicz. 2021. An Evaluation of Disentangled Representation Learning for Texts. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 1939–1951, Online. Association for Computational Linguistics.

Guojun Yin, Bin Liu, Lu Sheng, Nenghai Yu, Xiaogang Wang, and Jing Shao. 2019. Semantics disentangling for text-to-image generation. *Preprint*, arXiv:1904.01480.

A Visualization of Embedding Space

Visualization of the embedding spaces for the test set, showing the distribution and clustering of text embeddings generated by each of the five different model variants.



(a) Baseline

(b) Finetuned Default

(c) Finetuned Grouped

(d) Disentangled Default

(e) Disentangled Grouped

Figure 3: Embedding spaces of the test set for the text embeddings from the different models

B Text to Image Retrival Examples

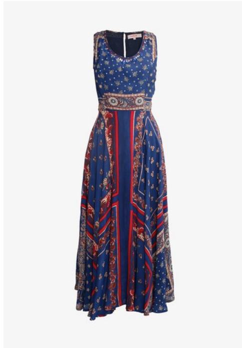
To qualitatively assess retrieval performance, we visualize the top-5 retrieved images for five random examples from the test set, comparing results across all model variants.

Query Text (German):

ein langes dunkelblaues gemustertes kleid, schulterfrei. der rock ist weit schwingend das mister in der farben rot und beige

Translation (English):

A long, dark blue patterned dress, off-the-shoulder. The skirt is wide and flowing. The dress is in red and beige.



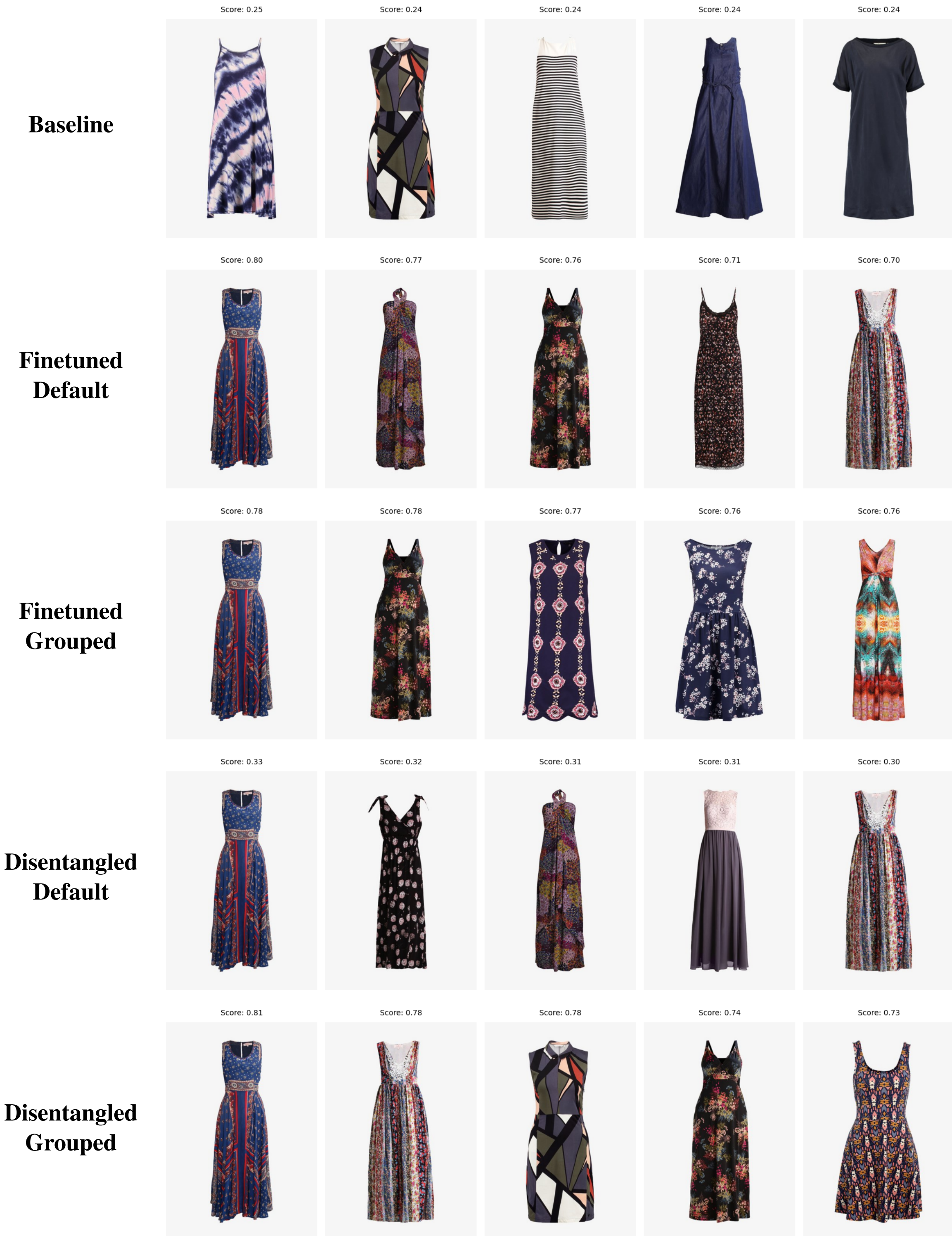


Figure 4: Qualitative retrieval results for five different model variants. Each row shows the top retrieved images for a given text query using a different model.

Figure 5: Qualitative retrieval results for five different model variants. Each row shows the top retrieved images for a given text query using a different model.

Query Text (German):

Ärmelloses enges Sommerkleid mit großem V-Ausschnitt. Das kurze Kleid ist aus Spitzenstoff.

Translation (English):

Sleeveless, fitted summer dress with a deep V-neck. The short dress is made of lace fabric.

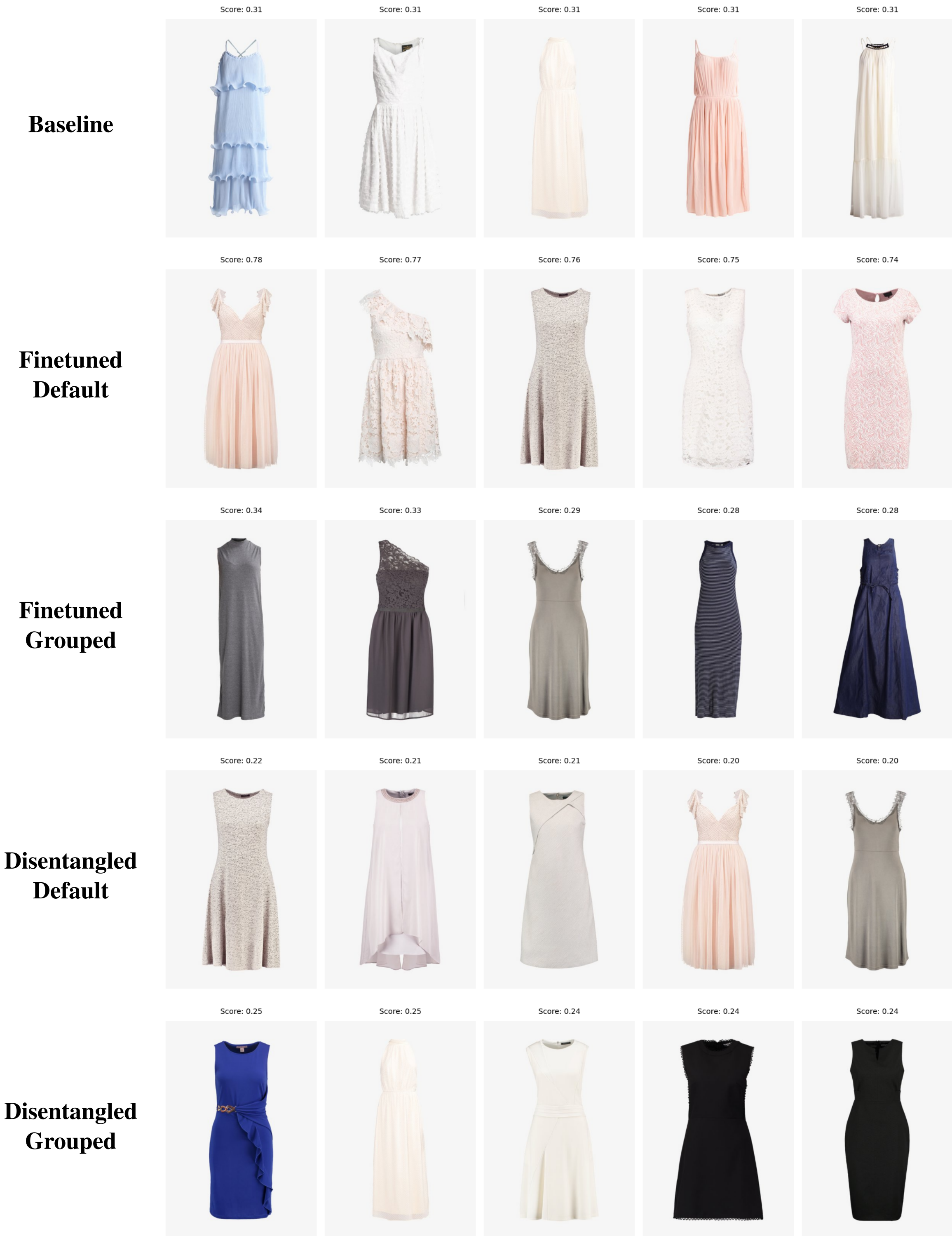


Figure 6: Qualitative retrieval results for five different model variants. Each row shows the top retrieved images for a given text query using a different model.

Query Text (German):

Knielanges Stoffkleid in rot mit Faltenrock, tailliert mit tiefem Ausshcnitt und schulterfrei.

Translation (English):

Knee-length red fabric dress with pleated skirt, fitted at the waist with a deep neckline and off-the-shoulder.

Figure 7: Qualitative retrieval results for five different model variants. Each row shows the top retrieved images for a given text query using a different model.

Query Text (German):

Ein kurzes, blaues, tailliertes Jeanskleid. Es ist ärmellos und hat einen eckigen Ausschnitt. Die Träger sind mit Knöpfen befestigt. Es hat eine Tasche im Brustbereich und zwei Taschen am Rock. Es hat vorne Knöpfe.

Translation (English):

A short, blue, fitted denim dress. It’s sleeveless with a square neckline. The straps are attached with buttons. It has one pocket on the chest and two pockets on the skirt. It has buttons at the front.

Figure 8: Qualitative retrieval results for five different model variants. Each row shows the top retrieved images for a given text query using a different model.

C Intra-Sample Distance Examples

This section presents detailed examples of intra-sample distance metrics across different model variants. For each item, we show the original German text descriptions with English translations, visualizations of the embedding space, and quantitative distance metrics. Lower distance values indicate that the model produces more consistent embeddings for semantically similar descriptions of the same garment.

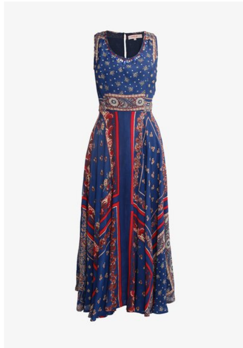
Example 1: Item 7914 (Blue Patterned Dress)

Model	**Data Type**	**Avg. Pairwise Distance**	**Max. Pairwise Distance**	**Mean Centroid Distance**
Baseline	Default	0.1449	0.2469	0.0599
Finetuned	Default	0.4771	0.9171	0.2138
Finetuned	Grouped	0.2040	0.4432	0.0852
Disentangled	Default	**0.0476**	**0.0942**	**0.0199**
Disentangled	Grouped	0.2275	0.4715	0.0958

Table 4: Intra-sample distance metrics for item 7914 (5 text embeddings, cosine metric). Lower values indicate more compact and consistent embeddings.

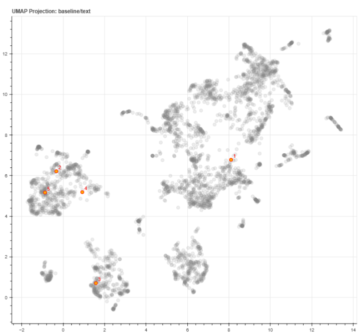
Query Text (German):

1. ein langes dunkelblaues gemustertes kleid, schulterfrei. der rock ist weit schwingend das muster in den farben rot und beige
2. Knöchellanges Stoffkleid in blau mit buntem Muster, weitem Faltenrock, tailliert, schulterfrei mit rundem Ausschnitt.
3. Ein dunkelblaues wadenlanges weites Kleid mit rotem Muster, einem großen V-Ausschnitt und ohne Ärmel.
4. Langes Sommerkleid ohne Ärmel. Das Kleid hat einen großen runden Ausschnitt. Das enge Oberteil hat an den Schultern attraktive Einsätze. Das Kleid ist blau mit teilweise Blümchen Druck und teilweise Bordüren Druck. Der Rock ist sehr weit.
5. Ein Kleid in überwiegend blauen Tönen. Es ist ärmellos und hat einen runden Ausschnitt.

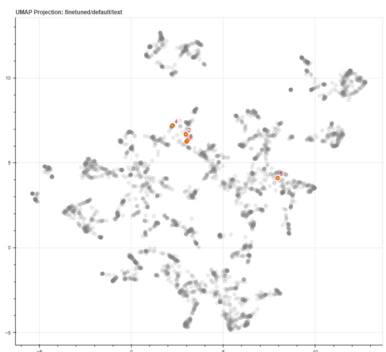


Translation (English):

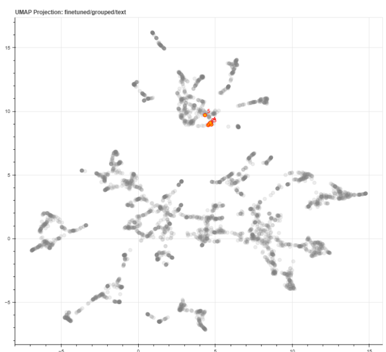
1. A long, dark blue patterned dress, off-the-shoulder. The skirt is wide and flowing. The pattern is in red and beige.
2. An ankle-length blue fabric dress with a colorful pattern, a wide pleated skirt, fitted at the waist, off-the-shoulder with a round neckline.
3. A dark blue, calf-length, wide dress with a red pattern, a deep V-neck, and no sleeves.
4. A long, sleeveless summer dress. The dress has a deep round neckline. The fitted bodice has attractive inserts at the shoulders. The dress is blue with a partial floral print and a partial border print. The skirt is very wide.
5. A dress in predominantly blue tones. It is sleeveless and has a round neckline.



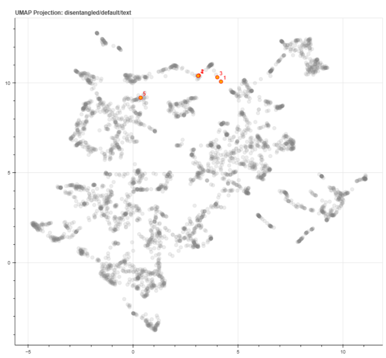
(a) Baseline



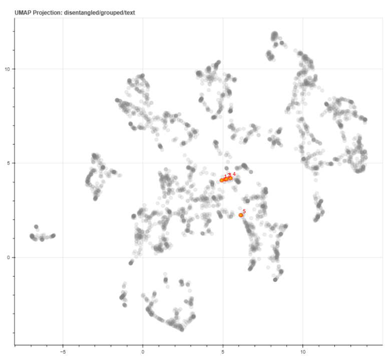
(b) Finetuned Default



(c) Finetuned Grouped



(d) Disentangled Default



(e) Disentangled Grouped

Figure 9: Embedding space visualization for item 7914 across different model variants. Each point represents one of the five text descriptions, colored by their position in the embedding space.

Example 2: Item 8442 (Blue Lace Dress)

Model	**Data Type**	**Avg. Pairwise Distance**	**Max. Pairwise Distance**	**Mean Centroid Distance**
Baseline	Default	0.0644	0.0970	0.0245
Finetuned	Default	0.2367	0.3720	0.0931
Finetuned	Grouped	0.7869	1.4200	0.3629
Disentangled	Default	**0.0289**	**0.0460**	**0.0109**
Disentangled	Grouped	0.5640	1.1323	0.2416

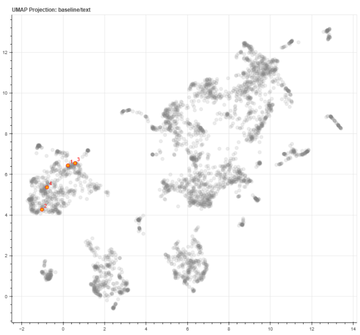
Table 5: Intra-sample distance metrics for item 8442 (4 text embeddings, cosine metric). Lower values indicate more compact and consistent embeddings.

Query Text (German):

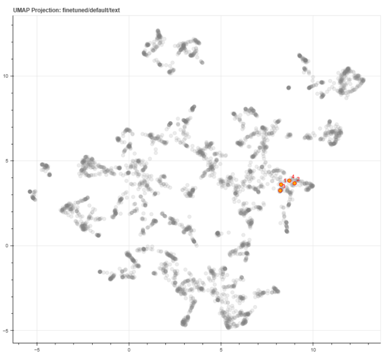
1. ein knielanges kleid mit blauem farbton. das kleid hat buntfalten im kleid
2. Ein dunkelblaues, knielanges Kleid. Das Oberteil ist aus Spitze. Der Rock ist aus Tüll. Das Kleid ist ärmellos, figurbetont, hat Spaghettiträger und einen eckigen Ausschnitt.
3. es ist ein blaues kleid es ist knietief und hat keine ärmel
4. fantasia Typ Kleid Blau mit viel Spitze, ohne Ärmel, nur zwei Riemen auf jeder Seite

Translation (English):

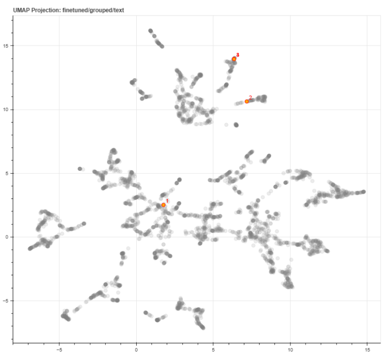
1. A knee-length dress in a blue hue. The dress has colorful pleats.
2. A dark blue, knee-length dress. The bodice is made of lace. The skirt is made of tulle. The dress is sleeveless, fitted, has spaghetti straps, and a square neckline.
3. It's a blue dress. It's knee-length and sleeveless.
4. A fantasia-style dress in blue with lots of lace, sleeveless, with only two straps on each side.



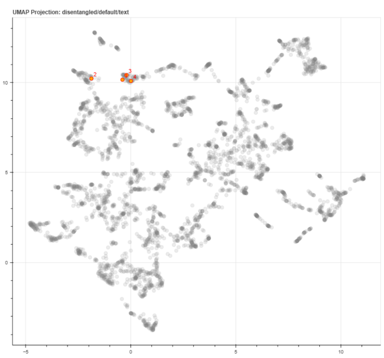
(a) Baseline



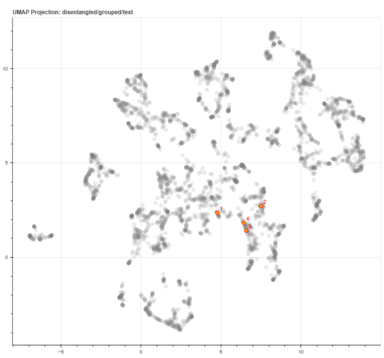
(b) Finetuned Default



(c) Finetuned Grouped



(d) Disentangled Default



(e) Disentangled Grouped

Figure 10: Embedding space visualization for item 8442 across different model variants. Each point represents one of the four text descriptions.

Example 3: Item 8564 (Red Lace Dress)

Model	**Data Type**	**Avg. Pairwise Distance**	**Max. Pairwise Distance**	**Mean Centroid Distance**
Baseline	Default	0.1227	0.1961	0.0504
Finetuned	Default	0.5951	1.0585	0.2769
Finetuned	Grouped	0.7089	1.0954	0.3421
Disentangled	Default	**0.0646**	**0.1133**	**0.0269**
Disentangled	Grouped	0.2800	0.6041	0.1193

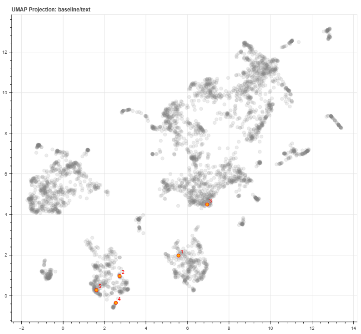
Table 6: Intra-sample distance metrics for item 8564 (5 text embeddings, cosine metric). Lower values indicate more compact and consistent embeddings.

Query Text (German):

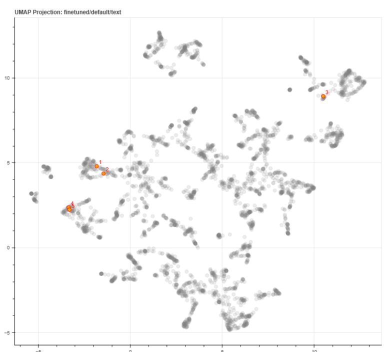
1. Ein kurzes, hell pinkes Kleid mit ausgeschnittenem Muster. Es besitzt einen tiefen V-Ausschnitt, keine Ärmel und die Schultern sind frei.
2. kurze schmal sexy elegantes rot kleid mit Spitze und ohne ärmel
3. Ärmelloses enges Sommerkleid mit großem V-Ausschnitt. Das kurze Kleid ist aus Spitzenstoff.
4. Knielanges und ärmelloses rotes Spitzenkleid. Es hat runden Ausschnitt. Perfekte Wahl für jeden Anlass.
5. Kurzes enges Kleid in rot. Es hat keine Ärmel und einen spitzen Ausschnitt.

Translation (English):

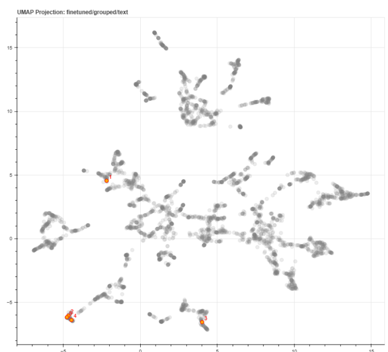
1. A short, light pink dress with a cut-out pattern. It has a deep V-neck, no sleeves, and bare shoulders.
2. A short, slim, sexy, elegant red dress with lace and no sleeves.
3. A sleeveless, fitted summer dress with a deep V-neck. The short dress is made of lace.
4. A knee-length, sleeveless red lace dress. It has a round neckline. Perfect for any occasion.
5. A short, fitted red dress. It has no sleeves and a pointed neckline.



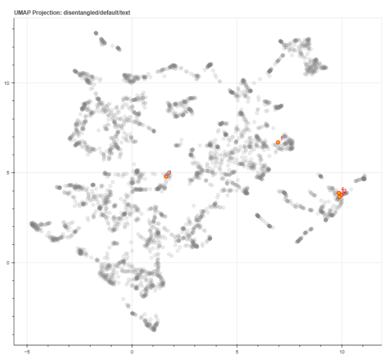
(a) Baseline



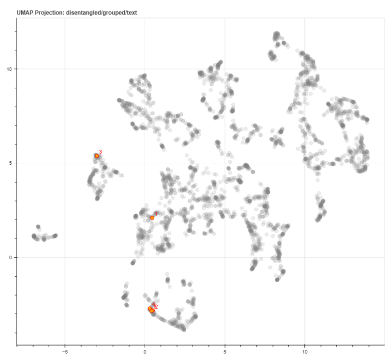
(b) Finetuned Default



(c) Finetuned Grouped



(d) Disentangled Default



(e) Disentangled Grouped

Figure 11: Embedding space visualization for item 8564 across different model variants. Each point represents one of the five text descriptions.

Example 4: Item 8681 (Pink Dress)

Model	Data Type	Avg. Pairwise Distance	Max. Pairwise Distance	Mean Centroid Distance
Baseline	Default	0.1342	0.2696	0.0553
Finetuned	Default	0.2694	0.5499	0.1143
Finetuned	Grouped	0.3601	0.8320	0.1562
Disentangled	Default	**0.0417**	**0.0906**	**0.0170**
Disentangled	Grouped	0.3572	0.7250	0.1554

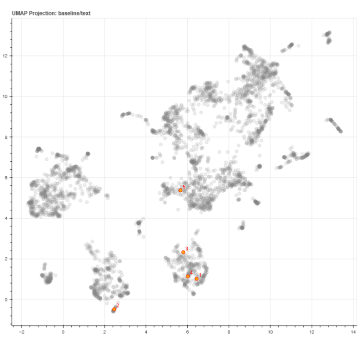
Table 7: Intra-sample distance metrics for item 8681 (5 text embeddings, cosine metric). Lower values indicate more compact and consistent embeddings.

Query Text (German):

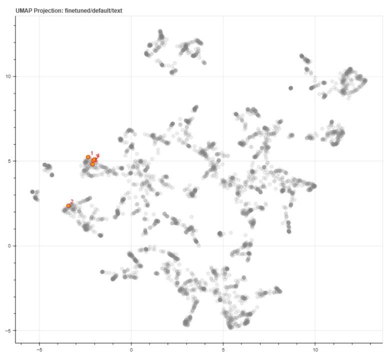
1. rosa rotes kleid es geht bis zu den Knien und hat einen breiten V-Ausschnitt es hat keine Ärmel
2. Knielanges Stoffkleid in rot mit Faltenrock, tailliert mit tiefem Ausschnitt und schulterfrei.
3. Ein pinkes Trägerkleid, welches bis oberhalb der Knie geht. Das Kleid hat einen kleinen Ausschnitt.
4. Ein kurzes, pinkfarbenes Kleid mit ausgestelltem Rock. Es hat keine Ärmel und einen kleinen V-Ausschnitt.
5. Hierbei handelt es sich um ein ärmelloses Sommerkleid mit einem rundlichen großen Ausschnitt in Altrosa. Das Kleid ist figurbetont und hat eine Länge bis zu den Unterschenkeln. Es eignet sich zum Ausgehen in Kombination mit einer kurzen Jacke, einer Umhängetasche sowie mit High Heels.

Translation (English):

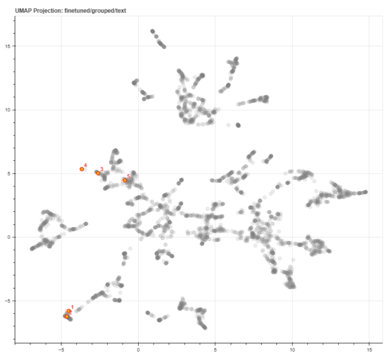
1. A pinkish-red dress that reaches the knees and has a wide V-neck. It is sleeveless.
2. A knee-length red fabric dress with a pleated skirt, fitted at the waist with a deep V-neck and off-the-shoulder design.
3. A pink sundress that reaches just above the knees. The dress has a small neckline.
4. A short, pink dress with a flared skirt. It is sleeveless and has a small V-neck.
5. This is a sleeveless summer dress with a rounded, wide neckline in dusty rose. The dress is fitted and reaches to the lower legs. It is suitable for going out when combined with a short jacket, a shoulder bag, and high heels.



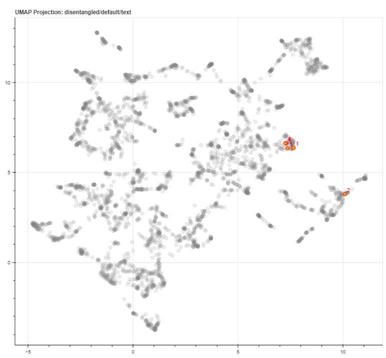
(a) Baseline



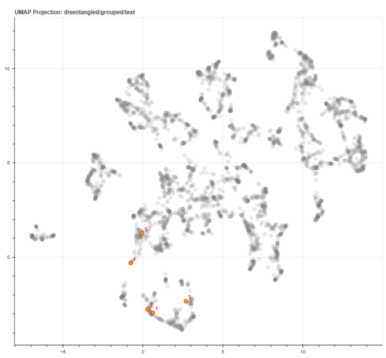
(b) Finetuned Default



(c) Finetuned Grouped



(d) Disentangled Default



(e) Disentangled Grouped

Figure 12: Embedding space visualization for item 8681 across different model variants. Each point represents one of the five text descriptions.

Example 5: Item 8581 (Denim Dress)

Model	Data Type	Avg. Pairwise Distance	Max. Pairwise Distance	Mean Centroid Distance
Baseline	Default	0.1928	0.3290	0.0667
Finetuned	Default	0.1308	0.2071	0.0446
Finetuned	Grouped	0.8832	1.3212	0.3588
Disentangled	Default	**0.0264**	**0.0388**	**0.0089**
Disentangled	Grouped	0.4972	0.7085	0.1884

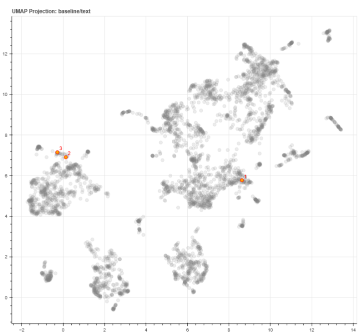
Table 8: Intra-sample distance metrics for item 8581 (3 text embeddings, cosine metric). Lower values indicate more compact and consistent embeddings.

Query Text (German):

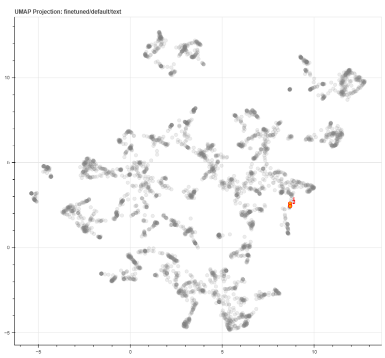
1. ein jeanskleid mit vielen Knöpfen vorne dran und Trägern. erinnert an latzhose
2. Ein kurzes, blaues, tailliertes Jeanskleid. Es ist ärmellos und hat einen eckigen Ausschnitt. Die Träger sind mit Knöpfen befestigt. Es hat eine Tasche im Brustbereich und zwei Taschen am Rock. Es hat vorne Knöpfe.
3. Ein Mittelblaues Jeans-ähnliches Minikleid, mit durchgehender Knopfleiste, drei Taschen vorn, geradem Ausschnitt, und schmalen Trägern.

Translation (English):

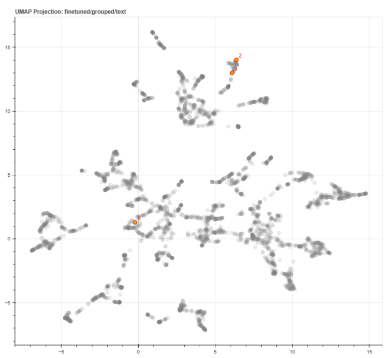
1. A denim dress with many buttons on the front and straps. It resembles overalls.
2. A short, blue, fitted denim dress. It is sleeveless and has a square neckline. The straps are attached with buttons. It has one pocket on the chest and two pockets on the skirt. It has buttons at the front.
3. A medium-blue, denim-like mini dress with a full button placket, three front pockets, a straight neckline, and narrow straps.



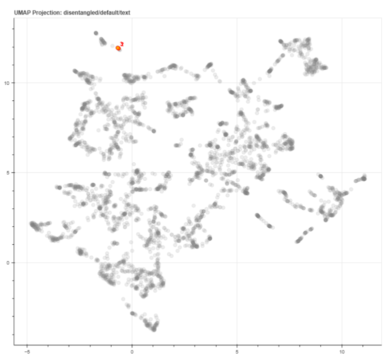
(a) Baseline



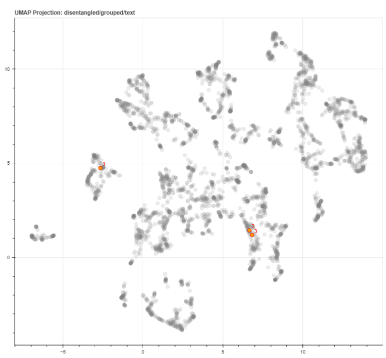
(b) Finetuned Default



(c) Finetuned Grouped



(d) Disentangled Default



(e) Disentangled Grouped

Figure 13: Embedding space visualization for item 8581 across different model variants. Each point represents one of the three text descriptions.